

Welcome to Code Crunch!

C7 | CRUNCH LABS: Wire Crunch - Embedded Systems

TUE 1-3PM Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



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Unit #2

C7 | Crunch LABS: Wire Crunch - Embedded Systems

Sensors and Data

What are sensors?

- They are devices that detect and respond to specific physical changes
 - Ex. Light, heat, pressure
- They are used in nearly everything
 - Ex. airbag systems, automatic doors, sensor pads
- The goal of sensor is to detect and respond accordingly and in a correct and safe manor

How Sensors work

- Detection - Sensors will detect a specific physical change
- Conversion - they convert the detected change to electrical signals
- Processing - the signals are processed by the microcontroller to extract the information needed
- Action - the information is relayed and the software designated for it is called upon and performs the action



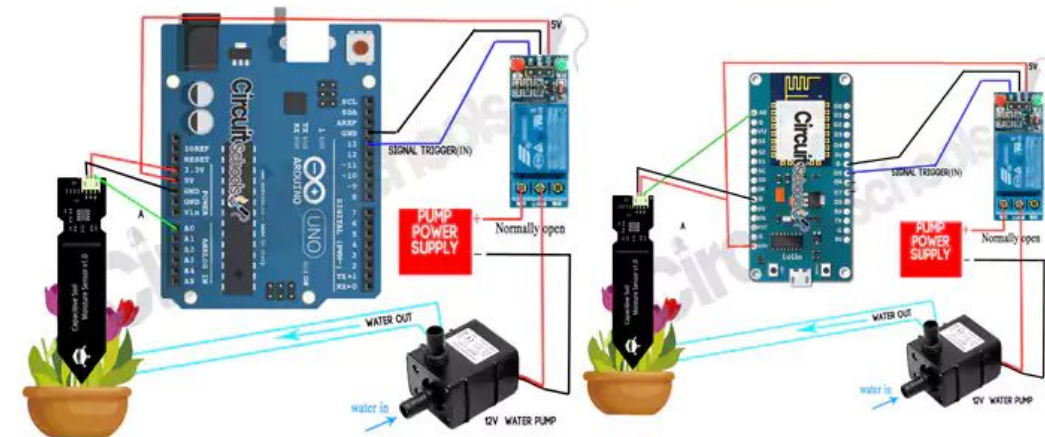
Reading Sensor values

- We use analog pins to read a sensors value
 - They fluctuate on a scale of 0-1023
- `analogRead(pin)` will output this value
- The value will tell you what is happening with the sensor
- Use Serial to print out the value that way we can understand it ourselves

Moisture sensors

- We want to read the value of the moisture sensor so we will use `analogRead(moisturePIN)`
- Moisture sensors measure the level of moisture in the soil
 - In arduino it measures the electrical resistance of the soil

AUTOMATIC PLANT WATERING SYSTEM USING ARDUINO OR ESP





Debugging out sensors

- First we want to make sure we are printing out our analog values with `analogRead`
- If the values are correct, we should include debug statements with `Serial.println()`
- Ensure your pins are correctly plugged in and declared in the setup
- If you were doing this physically you could use a voltmeter which measures the voltage in the electrical circuits
- Unit test each individual part to make sure functionality is working properly



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Q&A Session



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



RSVP lu.ma/CODECRUNCH



Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch